Journal of Sustainability for Energy
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Leveraging Artificial Intelligence for Enhanced Sustainable Energy Management

Swapandeep Kaur[1*], Raman Kumar[2*], Kanwardeep Singh[1], Yinglai Huang[3]

[1] Department of Electrical Engineering, Guru Nanak Dev Engineering College, 141006 Ludhiana, Punjab, India
[2] Department of Mechanical and Production Engineering, Guru Nanak Dev Engineering College, 141006 Ludhiana, Punjab, India
[3] College of Information and Computer Engineering, Northeast Forestry University, 150040 Harbin, China

[*] Correspondence: Swapandeep Kaur (ee_swapandeep@gndec.ac.in); Raman Kumar (sehgal91@gndec.ac.in)

Received: 11-23-2023 **Revised:** 01-05-2024 **Accepted:** 01-18-2024

Citation: S. Kaur, R. Kumar, K. Singh and Y. L. Huang, "Leveraging artificial intelligence for enhanced sustainable energy management," *J. Sustain. Energy*, vol. 3, no. 1, pp. 1–20, 2024. https://doi.org/10.56578/jse030101.

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Abstract: The integration of Artificial Intelligence (AI) into sustainable energy management presents a transformative opportunity to elevate the sustainability, reliability, and efficiency of energy systems. This article conducts an exhaustive analysis of the critical aspects concerning the AI-sustainable energy nexus, encompassing the challenges in technological integration and the facilitation of intelligent decision-making processes pivotal for sustainable energy frameworks. It is demonstrated that AI applications, ranging from optimization algorithms to predictive analytics, possess a revolutionary capacity to bolster intelligent decision-making in sustainable energy. However, this integration is not without its challenges, which span technological complexities and socio-economic impacts. The article underscores the imperative for deploying AI in a manner that is transparent, equitable, and inclusive. Best practices and solutions are proposed to navigate these challenges effectively. Additionally, the discourse extends to recent advancements in AI, including edge computing, quantum computing, and explainable AI, offering insights into the evolving landscape of sustainable energy. Future research directions are delineated, emphasizing the importance of enhancing explainability, mitigating bias, advancing privacy-preserving techniques, examining socio-economic ramifications, exploring models of human-AI collaboration, fortifying security measures, and evaluating the impact of emerging technologies. This comprehensive analysis aims to inform academics, practitioners, and policymakers, guiding the creation of a resilient and sustainable energy future.

Keywords: Artificial intelligence; Sustainable energy management; Optimization algorithms; Technological integration; Decision making; Emerging technologies; System resilience

1 Introduction

Given the growing environmental concerns and the need for sustainable practices, the energy industry is one of the key areas experiencing radical change. AI has become a major driver in transforming decision-making in sustainable energy management by developing advanced technologies [1]. The world's energy consumption has increased dramatically in the twenty-first century, calling for a paradigm shift in favor of environmentally benign and sustainable alternatives. Effective, environmentally friendly, and sustainable energy management is more critical than ever as traditional energy sources run out and worries about climate change grow [2, 3]. The dynamic issues presented by climatic unpredictability, resource constraints, and rising demand are beyond the capabilities of conventional energy planning and distribution approaches [4]. As a result, using AI has become a game-changing tactic to alter decision-making procedures and improve energy systems' sustainability, resilience, and efficiency [1, 5]. The historical development of sustainable energy practices highlights the turning points and difficulties influencing today's energy industry [6]. It highlights the vital need for creative decision-making methods. It allows a thorough analysis of AI technology's smooth integration into sustainable energy management frameworks [3, 7]. This paper examines the complex interactions between AI and decision-making theories in the context of sustainable energy, set against the background of the convergence of environmental imperatives and technological breakthroughs.

1.1 Objectives of the Review

The impetus for conducting this extensive assessment is the identification of a critical nexus between the revolutionary potential of AI and the growing issues in sustainable energy management [8, 9]. The need to provide practical and sustainable solutions is growing as the world's energy needs rise and environmental concerns deepen. The reasons behind investigating decision-making theories concerning incorporating AI into sustainable energy practices are explained [5, 10]. So, the multifaceted motivation for this review can be encapsulated in several key aspects.

- Addressing Complexity and Uncertainty: The dynamic and multifaceted context in which sustainable energy management functions is marked by uncertainties stemming from shifting technical landscapes, geopolitical considerations, and climatic variability. Conventional decision-making techniques frequently fail to offer flexible and responsive solutions for negotiating in this complex environment. The review seeks to clarify how AI, with its potential for data-driven analysis and machine learning skills, offers ways to address and reduce the uncertainties and complexities involved in making decisions about sustainable energy.
- Enhancing Efficiency and Resilience: Energy systems must be optimized for resilience and efficiency in addition to moving away from reliance on fossil fuels due to the pressing need to switch to sustainable energy. The potential for increasing energy production, delivery, and consumption efficiency exists with AI-driven decision-making. The review investigates and clarifies how AI technologies improve energy systems by increasing their resilience and ability to adjust to shifting operational and environmental situations.
- Aligning with Global Sustainability Goals: Sustainable energy practices are necessary in light of international agreements and promises to mitigate climate change and reduce carbon emissions. These practices should be in line with global sustainability goals. The review highlights how AI is crucial in attaining energy sustainability and making it easier for energy plans to align with general environmental and socio-economic goals.
- Fostering Innovation and Technological Integration: The use of AI in sustainable energy management is a cutting-edge research area that can alter preconceived notions completely. The study hopes to stimulate more investigation and cooperation between the scientific and business sectors by examining the driving forces underlying this integration. It aims to spark conversations about encouraging creativity and the smooth integration of cutting-edge technology for environmentally friendly energy solutions.

The motivations for investigating decision-making theories in the context of AI and sustainable energy management highlight the need for and possibility of revolutionary change in the quest for a more robust and sustainable energy future [8, 11]. Comprehending the review's breadth and establishing well-defined objectives are crucial elements that establish the parameters that will govern the investigation of decision-making theories about integrating AI and sustainable energy management. The key focus areas are a thorough examination of traditional and modern decision-making theories and how they may be applied and modified to meet the difficulties of sustainable energy management. This review aims to significantly contribute to the academic discourse and practical implementations at the nexus of AI and sustainable energy management decision-making by clearly stating its scope and objectives.

2 Foundations of Sustainable Energy Management

2.1 Principles of Sustainable Energy

The production, use, and conservation of energy resources are guided by a set of core concepts known as sustainable energy management, emphasizing social justice, economic viability, and long-term ecological balance. The fundamental ideas form the basis of the sustainable energy concept, providing a thorough grasp of the possibilities and problems associated with making decisions about sustainable energy management [12]. Optimizing energy efficiency and reducing waste is fundamental to sustainable energy. Adopting methods and technologies that maximize energy utilization at all production, distribution, and consumption phases are necessary. Sustainable energy management must go from using clean, renewable energy sources to limited and destructive ones for the environment [13]. These principles consider technological improvements, economic viability, and geographical fit. One essential idea is to recognize and minimize the effects of energy-related activities on the environment. Thorough environmental impact assessments are necessary for sustainable energy management decision-making to determine the energy projects' ecological footprint [14].

Sustainable energy management considers social and economic factors in addition to ecological ones. All stakeholders should be involved in decision-making processes, considering the socio-economic effects on communities, the equitable allocation of energy resources, and access to energy services. The ever-changing energy landscape and climate change uncertainties demand decision-making approaches that enhance system resilience and adaptation [15]. The contribution of decision-making theories to the design of energy systems that can resist shocks, adjust to changing circumstances, and guarantee a steady supply of energy. Adequate legislative and regulatory frameworks are critical to developing and applying sustainable energy measures. Making decisions entails negotiating intricate policy environments, predicting regulatory changes, and matching tactics with the government's goals [16].

2.2 Challenges in Sustainable Energy Management

Numerous technological, economic, social, and environmental obstacles confront sustainable energy management. Comprehending and tackling these obstacles is vital to creating efficacious decision-making approaches. It also lays the groundwork for future conferences on how decision-making theories, especially those enhanced by AI, can help find responses [17]. Table 1 shows challenges in sustainable energy management.

Table 1. Challenges in sustainable energy management

Challenges	Description	Impact on Decision-Making	Potential Solutions	Ref.
Intermittency of Renewable Sources	Variability in energy production from renewable sources, such as solar and wind, leads to intermittent power supply.	Decision-making must account for strategies to manage fluctuations and enhance reliability.	Advanced energy storage solutions, innovative grid technologies, and predictive analytics for better demand forecasting.	[9]
Technological Integration and Infrastructure	Challenges associated with integrating diverse, sustainable energy technologies into existing infrastructure.	Decision-making involves evaluating the compatibility, scalability, and interoperability of technologies.	Rigorous assessment of technological compatibility, phased implementation, and investment in scalable and interoperable systems.	[18]
Economic Viability and Affordability	Balancing the environmental benefits of sustainable energy with economic considerations to ensure project viability and affordability.	Decision-making must weigh economic feasibility and explore financing models for sustainable projects.	Cost-benefit analysis, financial incentives, and innovative financing mechanisms to enhance economic viability and affordability.	[18, 19]
Grid Management and Energy Storage	Efficient management of energy grids and the development of effective energy storage solutions to address supply-demand imbalances.	Decision-making involves optimizing energy distribution, storage capacities, and grid resilience.	They are integrating advanced energy storage technologies, demand-response strategies, and grid infrastructure upgrades.	[20]
Regulatory and Policy Uncertainties	The dynamic nature of energy policies and regulations introduces uncertainties impacting decision-making in sustainable energy projects.	Decision-making must navigate evolving legislative landscapes and ensure compliance with changing regulations.	Regularly monitor policy changes, proactive engagement with regulatory bodies, and develop adaptive strategies to comply with evolving regulations.	[21]
Community Engagement and Social Acceptance	The importance of gaining community acceptance and engagement in sustainable energy projects for their successful implementation.	Decision-making involves strategies for effective communication, stakeholder engagement, and addressing community concerns.	Community outreach programs, transparent communication, and incorporating local feedback into project planning and implementation.	[22, 23]

2.3 Importance of Decision-Making in Sustainability

In the field of energy management, sustainability is based on sound decision-making. Energy decisions significantly influence the environment, and making smart decisions is essential to reducing negative consequences. Reducing carbon emissions and protecting ecosystems are directly impacted by decisions to adopt sustainable practices, renewable technology, and resource-efficient procedures [1]. Moreover, the focus on conservation and resource efficiency highlights how crucial decision-making is to reducing waste production and resource depletion. Sustainable decision-making in this context supports the circular economy's tenets, including resource efficiency, reuse, and recycling, to promote long-term environmental sustainability [2, 20].

Decisions in the sustainable energy sector are inextricably tied to long-term profitability and economic resilience. Making well-informed decisions promotes resilience against economic shocks by balancing sustainability. The development of sustainable industries, the creation of jobs, and the general economic prosperity that comes with a switch to green energy are all significantly influenced by decision-makers [5, 24]. Decisions about sustainable energy should consider the various demands of communities to guarantee that everyone may benefit from clean energy [25]. Sustainable energy is supported, community involvement is increased, and energy poverty is decreased when inclusive decision-making methods are used [26].

Strategic decisions in the sustainable energy arena encourage technological innovation and progress. Decision-makers lead the industry to adopt new technology, support research and development, and continuously increase the effectiveness of energy solutions [27]. Maintaining a competitive edge and optimizing the overall performance of sustainable energy projects require this kind of creative dedication [28]. Finally, choices must align with regulatory frameworks and policies that support them. Policymakers negotiate intricate regulatory environments, guaranteeing adherence while promoting measures that encourage and promote sustainability. In sustainable energy management, decision-making is fundamentally important to sustainability since it contributes to larger economic growth objectives, social well-being, and environmental preservation [29, 30].

3 AI in Sustainable Energy Context

AI is leading the charge to transform sustainable energy management by providing solutions that use computational intelligence to solve difficult problems [31]. AI is, at its core, a collection of many technologies and approaches designed to allow robots to simulate human intellect [32]. A subset of AI called machine learning helps systems learn from and adapt to data, improving their capacity to carry out tasks without explicit programming [33, 34]. Computer vision enables computers to analyze and make decisions based on visual input, while natural language processing helps humans and machines communicate [35, 36]. Figure 1 depicts an overview of AI. It gives a basic introduction to the underlying ideas of AI, emphasizing its adaptability and versatility, which make it a potent instrument in the context of sustainable energy.

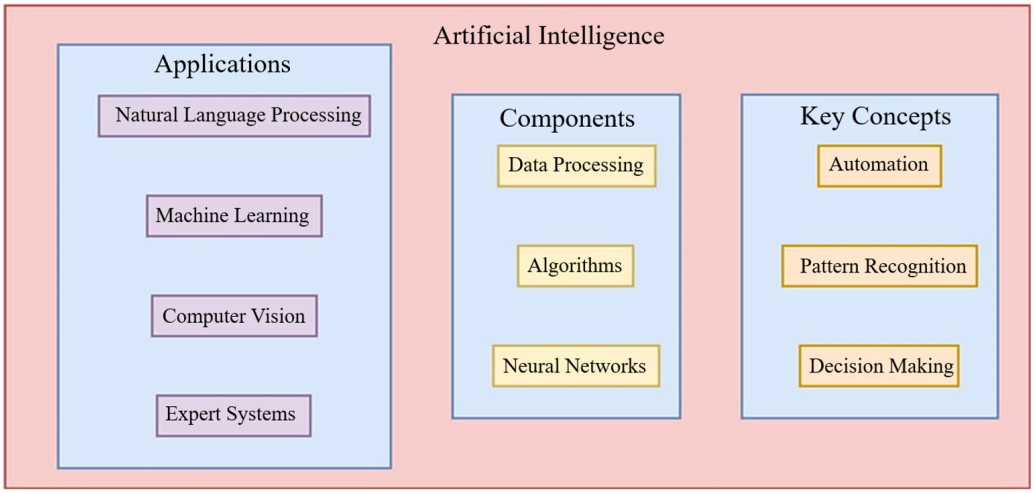


Figure 1. Overview of artificial intelligence

3.1 Applications of AI in Sustainable Energy

AI has several different revolutionary uses in sustainable energy [37]. AI technologies help optimize energy use, increase grid efficiency, and enable predictive maintenance in renewable energy infrastructure [38]. Machine learning algorithms make accurate energy demand possible by examining enormous databases [39]. AI-driven automation optimizes operational procedures, producing more effective energy systems [40]. Further detail on certain applications [41] shows how AI's automation, machine learning, and data analytics improve the sustainability and resilience of energy systems [2, 42]. Figure 2 presents applications of AI in sustainable energy.

3.1.1 Role of AI in decision-making processes

AI greatly enhances decision-making in sustainable energy management. AI's strong points are large-scale dataset processing, correlation analysis, and real-time prediction. AI aids in decision-making by anticipating energy consumption trends, optimizing resource allocation, and enabling proactive maintenance [2, 5]. The speed at which AI can collect and interpret data improves decision-makers efficacy and agility in navigating the intricacies of sustainable energy initiatives [43]. AI decision-making supports more thoughtful and strategic choices, promoting the overall sustainability of energy systems [1].

3.1.2 Smart technologies and the Internet of Things (IoT)

The interaction of AI, smart technologies, and the IoT brings about a paradigm shift in sustainable energy management. Energy-efficient appliances, smart grids, and Internet of Things-enabled sensors produce large volumes of real-time data. AI analyzes this data to produce insights that can be implemented, giving precise control over energy distribution, use, and storage [44]. Integrating AI with IoT and smart technologies promotes a comprehensive

and networked strategy for sustainable energy management. The synergy enables stakeholders to optimize energy systems, make data-driven decisions, and improve the energy landscape's sustainability and efficiency [45]. Figure 3 presents an overview of smart technologies and the IoT.

The application of AI in sustainable energy is not without difficulties and moral dilemmas despite its revolutionary promise. Data privacy, algorithmic biases, and openness in decision-making must be addressed to guarantee responsible and ethical use [46].

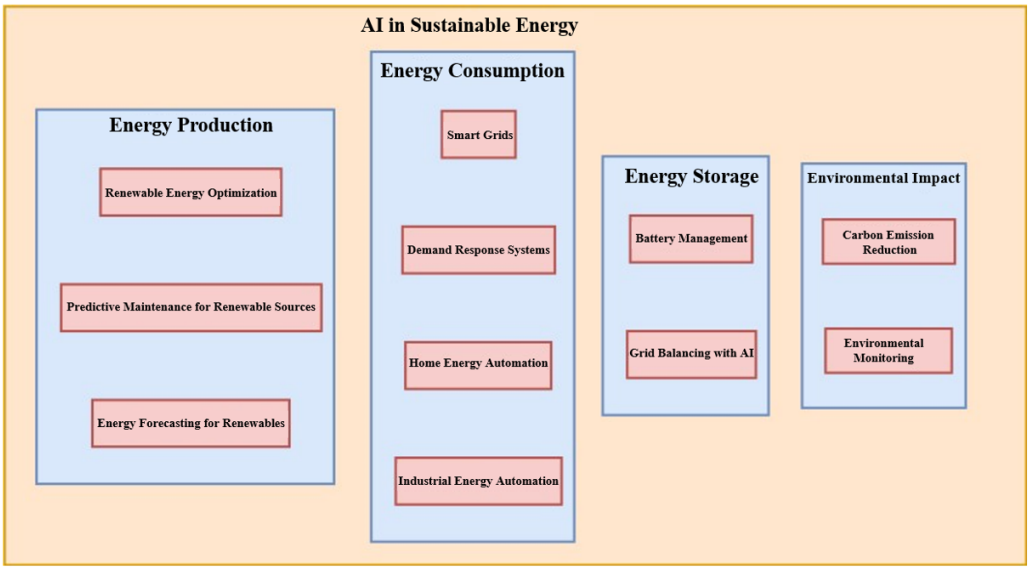


Figure 2. Applications of AI in sustainable energy

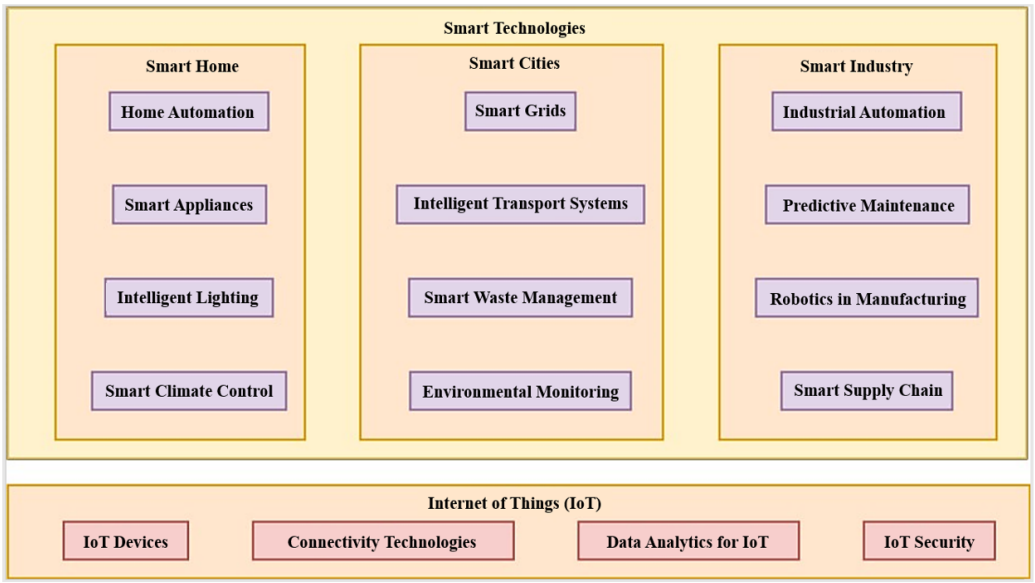


Figure 3. Overview of smart technologies and the Internet of Things

3.2 AI Techniques Relevant to Sustainable Energy

Table 2 thoroughly summarizes several AI approaches applicable to sustainable energy management. Each AI method is briefly explained, offering an understanding of its underlying ideas. The "Applications in Sustainable Energy" demonstrates the usefulness of these methods in real-world scenarios by showing how they support sustainable energy-related activities, including system control, energy optimization, and predictive analytics. The benefits and possible drawbacks of each AI method are outlined. Table 2 is an insightful overview of AI technologies for resolving issues and streamlining decision-making procedures in the ever-changing sustainable energy sector.

Table 2. AI techniques relevant to sustainable energy

AI Technique	Description	Applications in Sustainable Energy	Advantages and Considerations	Ref.
Machine Learning (ML)	Utilizes algorithms that enable systems to learn and improve from experience without explicit programming.	Predictive analytics for energy demand forecasting, optimization of energy consumption, and fault detection in energy systems.	Adaptive learning ability to analyze large datasets, but requires substantial training data.	[46, 47]
Neural Networks	Mimics the human brain's structure and processes to recognize patterns and make decisions.	Pattern recognition in energy consumption, optimization of grid operations, and predictive maintenance in renewable energy infrastructure.	Powerful for complex tasks but may lack transparency in decision-making.	[48, 49]
Natural Language Processing (NLP)	It enables machines to understand, interpret, and generate human-like language.	Chatbots for customer interactions, analysis of textual data in energy reports, and voice-activated control in smart energy systems.	Enhances human-machine communication may face challenges in understanding context and nuance.	[50, 51]
Computer Vision	It empowers machines to interpret and make decisions based on visual data.	Image recognition for monitoring energy infrastructure, visual inspection in maintenance processes, and security surveillance in energy facilities.	Enables automation in visual tasks but may require substantial computing resources.	[52, 53]
Reinforcement Learning	Involves an agent learning by interacting with an environment and receiving feedback in the form of rewards or penalties.	Control and optimization of energy systems, autonomous energy management in smart grids, energy-efficient building control.	It is well-suited for dynamic environments, but can be computationally intensive and challenging to train.	[54, 55]
Genetic Algorithms	Mimics the process of natural selection to find optimal solutions to complex problems.	Optimization of energy generation and distribution, design of energy-efficient systems, parameter tuning in algorithms.	Efficient for optimization problems, but may not guarantee finding the global optimum.	[56, 57]
Fuzzy Logic	Handles uncertainty by assigning degrees of truth to statements, allowing for more nuanced decision-making.	Energy demand forecasting, adaptive control in energy systems, decision-making in uncertain conditions.	Facilitates decision-making in ambiguous situations, but may require careful tuning of parameters.	[58]

3.2.1 Integration of AI in energy systems

Efficiency, resilience, and sustainability are all improved by the revolutionary paradigm represented by integrating AI into energy systems. AI technologies have become integral to the energy infrastructure, significantly contributing to operational optimization, demand forecasting, and resource management [59]. AI algorithms evaluate large datasets in the context of energy production and distribution, enabling predictive maintenance to reduce downtime, extend equipment life, and increase overall system dependability [60].

AI makes real-time monitoring, demand-response mechanisms, and grid optimization easier in smart grids. Due to machine learning algorithms, smart grids can adjust dynamically to fluctuating energy demands, which also helps them integrate renewable energy sources efficiently and balance the system for peak performance [61, 62]. Moreover, AI-powered energy storage management improves energy retrieval and storage, bolstering grid stability and offering vital assistance during instances of high demand [63].

The use of AI in energy systems greatly enhances decision-making. Smart algorithms, machine learning models, and advanced analytics enable stakeholders to make well-informed choices about energy generation, transmission, and distribution [64]. Precise forecasts are made possible by AI's capacity to process and evaluate a wide range of data sources, facilitating resource allocation and strategic planning [65]. Additionally, AI promotes energy

conservation by pointing out trends and abnormalities in use and directing people toward more environmentally friendly habits [66, 67].

AI integration in energy systems aims to advance wider sustainability objectives and improve operations. With the help of AI-driven insights, the energy landscape may become more robust, flexible, and ecologically friendly as we move toward cleaner, more renewable energy sources [68]. The smooth incorporation of AI into energy systems looks to be a key component in creating a sustainable and intelligent energy future [69, 70].

4 Decision-Making Theories and Models

Various frameworks, including theories and decision-making models, help people and organizations make well-informed decisions. These ideas offer methods to comprehend, evaluate, and choose the optimal action. Several well-known theories and models of decision-making are shown in Figure 4.

- **Rational Decision-Making Model:** Based on economic theory, this model believes that people make decisions by weighing their possibilities and selecting the one with the most predicted value. It entails a systematic and rational procedure prioritizing decision optimization and goal clarity [71, 72].
- **Bounded Rationality Model:** Herbert A. Simon's Bounded Rationality Model considers time, knowledge, and cognitive limits. Cognitive biases and decision-making play a role in sustainable energy analysis. This model has ramifications for several academic fields, including economics, psychology, and organizational behavior. It offers a more realistic explanation of human decision-making by considering the complexity and restrictions present in real-world circumstances [73, 74].
- **Behavioral Decision-Making Model:** This model investigates how psychological variables, emotions, and cognitive biases affect decision-making, drawing on behavioral economics and psychology. It emphasizes how people can deviate from rationality and become susceptible to prejudices such as anchoring, framing effects, and overconfidence [75, 76].
- **Garbage Can Model:** This model, which Cohen, March, and Olsen created, sees decision-making as a "garbage can" where issues, answers, and decision-makers come together randomly. Instead of following a systematic procedure, decisions are made on the spur of the moment, frequently reacting to unplanned circumstances or outside events [77, 78].
- **Cybernetic Model:** This model has its roots in the idea of feedback loops, which see decision-making as an ongoing process that involves obtaining data, making modifications in response to feedback, and refining decisions over time. It focuses on flexibility and learning from results [79, 80].
- **Political Decision-Making Model:** This paradigm, which acknowledges the impact of organizational politics, asserts that power struggles, coalition building, and negotiations occur within an organization to determine choices. It draws attention to stakeholders' roles and conflicting interests in influencing choices [81, 82].
- **Prospect Theory:** This hypothesis, which Kahneman and Tversky developed, shows that people often assess possible losses and profits asymmetrically, undermining the notion that people make rational decisions. It presents the idea of "loss aversion" and implies that people frequently base their decisions on how they perceive their reference point [83].
- **Game Theory:** Game theory examines decision-making in interactive scenarios where the result of one participant's choice depends on the decisions of other participants. It is extensively used in business planning, political science, and economics [84].

These decision-making theories and models provide a variety of perspectives to help in the understanding and enhancement of decisions. To negotiate difficulties and make wise decisions, decision-makers may use components from several theories, depending on the situation.

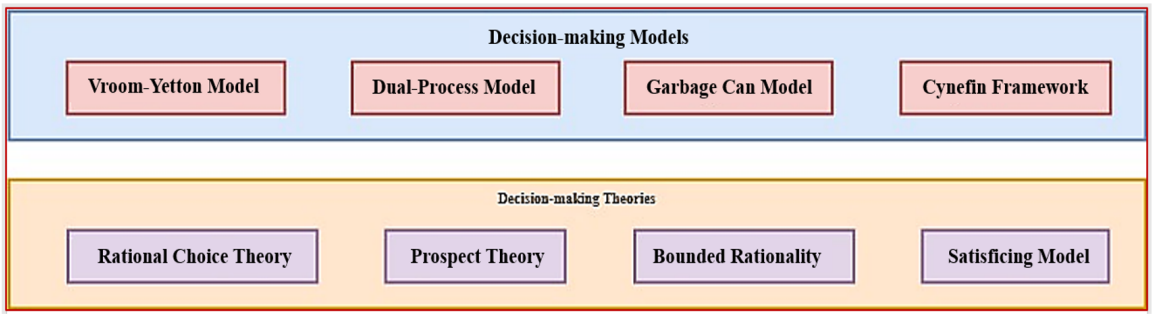


Figure 4. Decision-making theories and models

4.1 Classical Decision-Making Theories

The fundamental ideas of classical decision-making theories serve as a roadmap for rational decision-making for both people and organizations. These ideas use methodical and logical procedures to make the best choices. Here are a few well-known traditional theories of decision-making:

- **Rational Choice Theory:** This theory has its roots in classical economics, asserting that people should take decisions by comparing the advantages and disadvantages of many options and selecting the one that maximizes their expected utility. People are logical agents with distinct preferences who aim to maximize results and base their judgments on reliable and comprehensive information [85, 86].
- **Expected Utility Theory:** A framework for decision-making is provided by this theory. It integrates the probabilities of various outcomes and quantifies preferences in terms of value. This theory considers both the values and probabilities associated with possible outcomes. It provides a systematic method for simulating decisions under uncertainty and is widely used in game theory and economic decision-making. It helps academics and individuals comprehend and anticipate rational decision-making processes in various situations [87–89].
- **Decision Trees:** A visual depiction of choice alternatives, their outcomes, and their probability is offered by decision trees. This paradigm makes complicated choice scenarios easier to evaluate. It helps decision-makers evaluate the best options using branches and nodes to depict decision routes and uncertainty. Decision trees are frequently employed in risk analysis, project management, and operations research [90, 91].
- **Linear Programming:** Linear programming makes finding the optimal result in a linear mathematical model easier with linear programming, a mathematical technique for optimizing decision variables subject to restrictions. It involves following linear restrictions and either maximizing or minimizing an objective function. It is extensively employed in resource allocation, supply chain management, and operations research [92, 93].
- **Utility Theory:** The idea of utility, which stands for the arbitrary value or pleasure obtained from a result, extends the anticipated utility theory concept. It is utilized in research on consumer behavior, psychology, and economics [94, 95].

These traditional decision-making theories offer methods for evaluating options, considering preferences, and making the best judgments. Even though these theories presuppose a rational decision-making process, it's critical to understand that cognitive constraints, uncertainty, and contextual variables frequently cause real-world judgments to diverge.

4.2 Contemporary Decision-Making Models

Modern decision-making models integrate ideas from organizational behavior, sociology, and psychology to recognize the subtleties and complexity of the real world. The following are a few well-known modern models of decision-making:

- **Recognition-Primed Decision (RPD) Model:** The RPD model, created in the context of expert decision-making, suggests that seasoned decision-makers frequently spot patterns and instinctively select the most comfortable course of action without carefully weighing their options. Decision-makers use their experience, quickly pick up on clues, and instinctively connect patterns to well-solved problems in the past. It is especially pertinent in fast-paced, dynamic scenarios like military decision-making and emergency response [96, 97].
- **Naturalistic Decision Making (NDM) Model:** Like the RPD model, NDM involves decision-making in realistic, real-world contexts. It highlights how experience, instinct, and surroundings influence decisions. To make sound judgments, decision-makers must adapt to complicated and dynamic situations by drawing on their experience and contextual clues. It is utilized in fields like healthcare and aviation, where choices are made in spontaneous, realistic settings [98, 99].
- **Cognitive Continuum Theory:** This paradigm acknowledges a continuum of decision-making processes, ranging from analytical and logical processes to more intuitive and emotionally charged ones. Depending on the intricacy of the choice and the information at hand, decision-makers may alternate between analytical and intuitive modes [100, 101].
- **Adaptive Decision-Making Model:** The model acknowledges the dynamic character of choice settings and highlights decision-makers capacity to modify their approaches in response to feedback and evolving conditions. Throughout time, decision-makers hone their decision-making techniques by continually learning from results and modifying their mental models. It is used in areas like company strategy and innovation, where choices are influenced by feedback and uncertainty [102, 103].
- **Vroom-Yetton Decision Model:** This model offers a decision tree that helps managers choose a decision-making approach depending on the type of choice and the degree of involvement from their staff. It assists leaders in decision-making processes. It is utilized to enhance decision-making processes in organizational studies and leadership [104, 105].

The dynamic, context-dependent character of choice settings and the inclusion of behavioral and cognitive elements in the decision-making process are now acknowledged in these modern decision-making models. They provide insightful guidance for negotiating the intricacies of contemporary decision-making situations.

4.3 Decision Making: Relevance to Sustainable Energy Management

Modern decision-making models are essential in sustainable energy management, where intricate and multifaceted issues need flexible and well-informed decision-making procedures. Using these models is critical to managing the unique dynamics of the energy sector's shift towards sustainability.

- **Recognition-Primed Decision (RPD) Model:** The RPD model can be useful for professionals in the quickly changing field of renewable energy, where unexpected obstacles may arise. In instances like renewable energy integration and grid resilience, recognizing patterns and trusting intuition become essential, enabling decision-makers to leverage their knowledge to traverse difficult situations effectively [99].
- **Naturalistic Decision Making (NDM) Model:** NDM fits nicely with the unpredictability of sustainable energy scenarios. Deploying renewable energy and responding to environmental repercussions are two areas where decision-makers must adjust to the intricacies of the actual world. In these dynamic circumstances, NDM offers a framework for utilizing experience and contextual clues to make wise judgments [98, 99].
- **Cognitive Continuum Theory:** The notion of cognitive continuum holds significance in comprehending the many methods of decision-making involved in sustainable energy management. To meet a variety of issues, decision-makers in the sustainable energy industry navigate a spectrum of cognitive processes, from analytical considerations in policy formation to intuitive judgments in community participation [101].
- **Adaptive Decision-Making Model:** In light of the fast advancement of sustainable energy policies and technology, decision-makers must adjust to evolving situations and new knowledge. The iterative and learning-oriented strategy needed in domains like energy storage implementation, where feedback loops are critical for gradually improving tactics, aligns with the adaptive decision-making paradigm [102, 103].
- **Sensemaking Model:** In sustainable energy management, sensemaking is crucial, especially when handling contentious or confusing topics like community support for renewable energy projects. Sensemaking is a tool that decision-makers may use to understand stakeholder viewpoints, formulate arguments persuasively, and reach judgments that support sustainability objectives [106, 107].
- **Vroom-Yetton Decision Model:** The Vroom-Yetton model may be utilized by leaders in sustainable energy firms to ascertain the most suitable decision-making approach, considering the particular circumstances. For example, including stakeholders in community solar project decision-making promotes collaborative decision-making and increases support [104, 105].

These modern models give sustainable energy a more comprehensive grasp of decision-making procedures. Acknowledging the usefulness of these models enables professionals to implement more sensible and efficient decision-making techniques, which helps the energy industry successfully navigate sustainability difficulties.

5 Applications of AI in Sustainable Energy Decision-Making

5.1 Predictive Analytics for Energy Consumption

To tackle the intricacies of contemporary energy systems, AI is essential in sustainable energy decision-making. Predictive analytics, which uses AI technology and sophisticated algorithms to anticipate future energy usage patterns, is one important application of AI. Using historical and current data, predictive analytics builds models that project future patterns in energy use [108]. To provide precise forecasts, AI systems, particularly machine learning models, examine massive datasets that include meteorological trends, the time of day, and past usage [109]. Proactive energy management requires this application since it helps stakeholders predict changes in demand and allocate resources as efficiently as possible [110].

By predicting times of peak demand, predictive analytics helps utilities and energy suppliers arrange resources more efficiently. This lowers the possibility of shortages or overproduction by ensuring enough energy generation and distribution capacity [111, 112]. Companies and individuals may modify their energy usage habits by forecasting spikes and valleys in energy demand.

Predictive analytics is integrated into smart grid systems to estimate energy consumption. This allows grid operators to balance supply and demand in real time. Predictive analytics powered by AI helps demand response programs, encouraging users to reduce their energy use during peak hours to maintain grid stability. Predictive analytics for sustainable energy decision-making will become more accurate by combining data from various sources, such as weather prediction models and the Internet of Things [113, 114]. The development of machine learning algorithms will continue to yield increasingly complex insights into energy usage patterns and facilitate proactive decision-making for a robust and sustainable energy infrastructure [109, 110].

5.2 Optimization Algorithms for Energy Distribution

AI is being applied to optimize energy distribution in the context of sustainable energy decision-making. AI-powered optimization techniques are essential for improving the dependability and efficiency of energy distribution networks [115, 116].

Optimization algorithms use AI methods like machine learning and mathematical modeling for energy distribution to expedite the routing and distribution of energy inside a grid [117]. These algorithms are designed to reduce energy losses, maximize the use of renewable resources, and improve distribution networks' overall efficiency. The best routes for distributing energy may be found through optimization algorithms that reduce transmission losses and raise the grid's overall effectiveness [118]. Optimization algorithms assist in balancing the intermittent characteristics of renewable energy sources, such as wind and solar power, by shrewdly storing and channeling energy within the distribution network. These algorithms support grid resilience by foreseeing possible interruptions and dynamically rearranging energy distribution paths, guaranteeing a consistent supply despite difficulties [4, 119].

Utilities employ optimization algorithms to ascertain the optimal paths for energy distribution, considering variables including load, distance, and network limitations. Localized sustainability is enhanced by optimization algorithms in microgrid systems, which guarantee effective energy distribution among distributed energy supplies, storage systems, and demand centers [120]. Future energy distribution optimization algorithms will use real-time data from sensors, IoT devices, and smart grids. Developing machine learning models will enable the growth of adaptive optimization techniques that can react quickly to shifting circumstances [121]. Moreover, blockchain technology might improve energy transactions' security and transparency in dispersed networks [122].

5.3 AI-Driven Resource Allocation in Sustainable Energy Projects

The application of AI significantly enhances resource allocation optimization for sustainable energy projects. AI uses sophisticated algorithms and data analytics to enhance resource efficiency, cost-effectiveness, and project success. The intelligent distribution of human, financial, and technical resources to optimize the efficiency and sustainability of energy projects is known as AI-driven resource allocation. This procedure makes data-driven judgments about resource allocation by analyzing many parameters using machine learning algorithms and optimization approaches [123].

AI algorithms examine past project data and market trends to improve budget allocation. This ensures that resources are used effectively to maximize the return on investment. By considering a variety of parameters, machine learning models can evaluate project risks, allowing for the proactive detection and mitigation of any issues before they become more serious. By considering resource availability, possible bottlenecks, and dependencies, AI aids in the creation of realistic project timetables that ensure the timely completion of sustainable energy efforts.

AI evaluates environmental and geographic aspects to optimize energy production and determine the best location for renewable energy installations, such as wind turbines or solar panels. To optimize resource allocation for building energy storage systems and improve grid stability, AI algorithms examine patterns of energy usage and storage technology [124]. AI will increasingly integrate real-time data from sensors, IoT devices, and smart grids to allocate resources to sustainable energy projects. Advanced predictive analytics will improve resource allocation models' accuracy, enabling dynamic modifications in response to shifting project conditions. Furthermore, resource allocation for intricate sustainable energy projects will be streamlined using AI to automate decision-making procedures.

6 Integration Challenges and Solutions

6.1 Technical Challenges

To fully realize the potential of these technologies, several problems related to the technological integration of AI into sustainable energy systems must be carefully considered. Managing a variety of data sources with differing quality is one of the main challenges [121]. Securing consistency, precision, and entirety in data requires strong governance plans, standardization procedures, and ongoing data quality evaluations [125]. Another challenge is interoperability with current technologies and systems, which necessitates the creation of open-source, modular designs and standardized interfaces that enable smooth communication between various components.

Significant hurdles arise regarding scalability and performance optimization, especially when expanding AI applications to meet the needs of large-scale energy systems. Optimizing algorithms and implementing distributed computing architectures are crucial to guaranteeing effective performance in growing energy systems [126]. Due to the overwhelming security and privacy concerns, strong cybersecurity methods, encryption protocols, and privacy-preserving AI algorithms must protect sensitive energy data. In situations where transparent decision-making procedures are crucial, the explainability and transparency of AI judgments assume a crucial role [127]. The energy efficiency of AI models also has to be addressed, emphasizing the creation of energy-efficient algorithms, investigating possibilities for edge computing, and putting strategies like model pruning and compression into practice [128]. Moreover, the capacity of AI models to adjust to changing environmental circumstances is essential

to their performance in renewable energy systems [129]. It is ensured that AI systems can adapt to changing conditions by being designed with methods for constant retraining and adaptive learning [130].

6.2 Ethical and Social Challenges

AI integration in sustainable energy systems raises several ethical and societal issues that must be carefully considered [131]. It is critical to ensure ethical and fair deployment, examining the consequences of AI technologies as they become increasingly integrated into decision-making processes and influence the energy landscape [132].

Among the main ethical concerns are accountability and transparency in AI decision-making. Transparency techniques are becoming increasingly important as AI models and incredibly sophisticated deep learning algorithms are frequently perceived as impenetrable "black boxes" [133].

There are important societal ramifications from AI systems' propensity for prejudice. AI systems can unintentionally reinforce and even worsen preexisting biases in training data if they are not properly built and maintained [134]. This raises fairness, equity, and the possibility of biased results [135]. Addressing bias necessitates paying close attention to dataset selection, algorithm design, and continuous monitoring to correct and prevent biased decision results.

Privacy issues can come into play, particularly when sensitive energy data is involved. It's critical to balance safeguarding individual privacy with using AI for data-driven insights [136]. Safeguarding sensitive personal data requires privacy-preserving AI algorithms, strict adherence to privacy legislation, and robust data anonymization processes [137]. A thorough analysis of AI's socio-economic effects on sustainable energy is needed. Although AI can potentially improve sustainability and energy efficiency, it may also result in job displacement in some industries. Proactive steps like reskilling and upskilling programs are needed to address the social ramifications and provide a fair transition for workers impacted by AI-driven changes in the energy profession [138].

It's also important to pay attention to issues like the price and accessibility of AI-driven technologies. It's crucial to ensure that AI's advantages in sustainable energy are spread fairly among all socio-economic classes and geographical areas to avoid worsening existing inequalities. A collaborative and interdisciplinary approach is essential in negotiating these ethical and societal issues [139]. To set ethical norms, implement responsible practices, and build an inclusive conversation about the social consequences of AI in sustainable energy, it is imperative to engage with various stakeholders, including ethicists, community leaders, politicians, and AI developers. This strategy aids in creating moral frameworks that place a high value on openness, justice, and the welfare of society [130, 134].

6.3 Proposed Solutions and Best Practices

It takes a proactive and moral approach to address the ethical and social ramifications of incorporating AI into sustainable energy systems. By implementing suggested solutions and best practices, stakeholders may promote responsible AI deployment, encourage transparency, and lessen potential biases [120, 140].

- **Transparency and Explainability:** Provide AI models with integrated transparency features and ensure stakeholders understand the decision-making process. Employ interpretable AI models, provide transparent documentation of decision-making procedures, and provide intuitive user interfaces that convey AI-driven insights understandably.
- **Bias Mitigation:** Act to find and fix AI algorithmic biases to ensure fairness and equitable results. Diversify training datasets to reduce preexisting prejudices. Audit and check AI systems often for biases and include a variety of stakeholders in the development process to offer a range of viewpoints.
- **Privacy Protection:** Use AI approaches that protect privacy to protect sensitive energy usage data and comply with privacy laws. To detect and reduce privacy concerns, put strong data anonymization procedures into place, encrypt data while it's in transit and at rest, and carry out privacy impact assessments.
- **Socio-Economic Impact Mitigation:** Establish programs and policies to deal with possible job displacement, ensuring that people affected by changes led by AI have a fair transition. Create initiatives for reskilling and upskilling, interact with educational institutions, and encourage diversity in the workforce to lessen adverse socio-economic effects and advance inclusive economic growth.
- **Equitable Access and Affordability:** Encourage programs that make AI's advantages for sustainable energy affordable and available to various socio-economic levels. To avoid making already-existing gaps worse, prioritize inclusion in the deployment of AI, actively include communities in decision-making processes, and encourage open access to AI-driven tools and technology.
- **Multidisciplinary Collaboration:** To address ethical and societal issues, promote cooperative efforts between ethicists, community leaders, legislators, and AI developers. Create multidisciplinary working groups and lead continuing discussions to integrate different viewpoints and promote group decision-making.
- **Continuous Evaluation and Adaptation:** Provide procedures for assessing and modifying AI systems to handle new moral and societal issues. Establish feedback loops for ongoing learning, evaluate the social effects of AI in sustainable energy regularly, and modify practices and regulations in response to changing ethical concerns.

By implementing these suggested solutions and best practices, stakeholders may proactively manage the ethical and social issues of incorporating AI into sustainable energy systems. This strategy helps establish trust, promotes ethical AI research, and guarantees the advantages of AI-powered innovations.

6.4 Future Trends and Emerging Technologies

The subject of AI integration into sustainable energy systems is dynamic, and several upcoming trends and cutting-edge technologies will likely influence its direction. These developments might improve energy systems' sustainability, resilience, and efficiency.

- **Explainable AI (XAI):** More focus is being placed on creating AI models with improved explainability to address concerns about transparency. It enhanced comprehension of AI-driven choices, encouraging stakeholder confidence and streamlining regulatory compliance.
- **Edge AI:** Edge computing is becoming more widely used for AI applications, allowing for real-time processing and decision-making at the data source. Edge device autonomy, efficiency, and latency have all grown, making AI more adaptable to changing energy system conditions.
- **Federated Learning (FL):** FL growth in the application of federated learning strategies enables training AI models on dispersed devices without the need for raw data exchange.
- **Exascale Computing:** Exascale computing and other developments in high-performance computing make it possible to process enormous volumes of data for intricate AI models. There is enhanced simulation performance, quicker model training, and the capacity to manage complex energy system dynamics.
- **Blockchain Integration:** There is increasing research into using blockchain technology to improve the security and openness of data sharing and transactions powered by AI. It enhanced the traceability of AI model outputs, led to safer energy transactions, and increased confidence in data integrity.
- **Neuromorphic Computing:** Growing interest in neuromorphic computer architectures due to inspiration from the composition and capabilities of the human brain leads to improved energy efficiency and the possibility of creating AI models with cognitive capacities similar to those of humans.
- **Generative Adversarial Networks (GANs):** GANs should be developed and used more often to create AI models and increase their resilience. GAN increased the diversity of training data, resolution of bias issues, and improvement of AI models' performance in various dynamic energy scenarios.
- **Exponential Growth in Data Analytics:** There is an unprecedented expansion in the quantity and diversity of data accessible for AI applications due to the widespread use of IoT devices and sensors. This also leads to enhanced predictive analytics, a deeper understanding of energy usage trends, and better decision-making for sustainable energy management.
- **Human-AI Collaboration:** There is a growing emphasis on collaborative frameworks in which AI technologies assist human specialists in making decisions. Synergistic capabilities and making moral and effective decisions are possible by fusing human intuition and contextual knowledge with AI's analytical prowess.
- **Quantum Computing:** Quantum Computing investigates and helps to resolve challenging optimization issues pertinent to the design and administration of energy systems. With further development, these trends and technologies can enhance AI's capacity for sustainable energy, opening the door to creative solutions, enhanced system resilience, and a more sustainable energy future. To fully utilize these advancements' revolutionary potential for the good of society and the environment, stakeholders in the area should watch them carefully.

7 Conclusions

AI in sustainable energy management is a cutting-edge technical innovation that has the potential to change our understanding of urgent energy-related issues completely. AI's revolutionary powers, from optimization algorithms to predictive analytics, provide hitherto unseen possibilities for improving the sustainability and efficiency of energy systems. However, this is not an easy road since technological integration issues, moral dilemmas, and societal ramifications must be carefully considered.

Strong solutions are required for the technological integration problems, which range from different data sources to scalability issues. These solutions include standardized interfaces, privacy-preserving methods, and creating energy-efficient models. Furthermore, to deal with the moral and social effects, we need a multifaceted plan that includes programs to lessen the negative social and economic effects, AI models that are clear and easy to understand, and methods to reduce bias. Future-focused trends and technologies like edge computing, federated learning, and quantum computing present intriguing opportunities. Cooperative efforts and ongoing appraisal are essential to exploit these developments appropriately and steer towards a more robust and sustainable energy future.

7.1 Implications for Future Research

The investigation of AI in sustainable energy management opens up several research directions that have the potential to significantly improve our comprehension and use of these technologies in the future. Subsequent studies should focus on creating sophisticated explainability methods for AI models related to sustainable energy. It is important to comprehend complex algorithms and make them more transparent, especially regarding important decision-making processes. Investigating new approaches for real-time bias prevention and detection in AI models functioning in dynamic energy contexts is imperative. Research should focus on adaptive techniques that continually address biases arising in dynamic datasets. This involves investigating novel approaches to safe data exchange, improving federated learning, and creating uniform frameworks for privacy effect assessments.

Future studies should focus on comprehensive socio-economic effect analyses of AI integration in sustainable energy. To ensure that the advantages are spread fairly, this entails comprehending the repercussions on employment, economic institutions, and community dynamics. Research on efficient models for AI systems and human specialists working together on sustainable energy decision-making is desirable. The system's performance may be improved by knowing how to best combine AI analytics with human intuition and experience. Strengthening the security and resilience of AI applications in the energy sector should be the main focus of research. Collaboration in cross-disciplinary research is crucial. Experts from various disciplines, including AI, energy engineering, ethics, and the social sciences, should be involved in future research.

Research should evaluate the unique uses and effects of new technologies, such as quantum computing and neuromorphic computing, on sustainable energy systems as they become more widely used. For these technologies to be successfully integrated, it is essential to comprehend how to use them properly. Evaluating the long-term robustness of sustainable energy systems with AI integration should be the main focus of research. This entails researching how AI models adjust to shifting energy environments, technological advancements, and unanticipated disruptions. Future studies in these areas will help AI technologies in sustainable energy mature and promote ethical, inclusive, and significant solutions. Through examining these ramifications, scholars may assume a crucial function in molding the trajectory of sustainable energy management, guaranteeing its efficiency, equity, and durability.

Author Contributions

Conceptualization, R.K.; methodology, K.D.; software, R.K.; validation, R.K. and S.K.; formal analysis, S.K.; investigation, R.K.; resources, R.K.; data curation, S.K.; writing—original draft preparation, K.D.; writing—review and editing, S.K., K.D.; visualization, I.M.; supervision, R.K.; project administration, S.K.; funding acquisition, Y.Y. All authors have read and agreed to the published version of the manuscript.

Data Availability

The experimental data used to support the findings of this study are available from the author upon request.

Acknowledgements

The authors gratefully acknowledge the generous support extended by Guru Nanak Dev Engineering College, Ludhiana, Punjab, and IKG Punjab Technical University, Kapurthala Jalandhar, which significantly contributed to the successful completion of this research.

Conflicts of Interest

The authors declare no conflicts of interest regarding this work.

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